

Yaohui Shu

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Ph.D. candidate in Applied Mathematics and Statistics working across computational applied mathematics, stochastic modeling, information-theoretic causal inference and turbulent processes in complex systems. The dissertation develops entropy-based methods for causality analysis, verified on physical models of turbulence-microphysics interactions with known causal structure.

EDUCATION

Stony Brook University, Ph.D., Applied Mathematics and Statistics, GPA 3.78/4.0 Expected May 2027

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

University of Rochester, M.S., Finance, GPA 3.81/4.0; Certificate in Quantitative Finance (CQF) Aug 2021 – Aug 2022

Undergraduate: B.A. Finance, Zhongnan Univ. of Economics and Law, 2021; B.S.B.A., Univ. of Rhode Island, 2020 (GPA 3.98)

RESEARCH EXPERIENCE

Stony Brook University Stony Brook, NY

Doctoral Researcher: Information-Theoretic Causal Inference for Complex Systems Aug 2023 – Present

- Formulated causal discovery without acyclicity or causal sufficiency: a partial ancestral graph (PAG) over a lag window under sigma-separation, with nearest-neighbor conditional mutual information tests and orientation from nonstationarity.
- Proposed a time-resolved causal strength that stays defined under feedback: the relative entropy rate between path measures with and without one drift dependence, in closed form by Girsanov's theorem.
- Treated unmeasured drivers as a measurement model, so the nonstationarity used for orientation also breaks the rotation indeterminacy of latent factors; added a surrogate test separating nonlinearity from nonstationarity.
- Built an installable Python package that embeds entropy-based measures (transfer entropy, conditional mutual information, information flow) in causal graph learning, tested against closed-form results and run in parallel on a SLURM cluster.
- Early-stage application to financial markets (in progress): PAG search with hidden drivers, change-point detection, latent factors and event studies on cross-asset, stock and options data, checked out of sample.

Brookhaven National Laboratory (U.S. Department of Energy)

Upton, NY

Graduate Research Assistant: Stochastic Modeling and Information Flow

Jan 2024 – Jan 2026

- Sole Ph.D. researcher on a DOE-funded cloud turbulence program: set the direction and built the theory and code.
- Generalized the McGraw-Liu drift-diffusion model to a coupled system driven by an Ornstein-Uhlenbeck saturation process spanning the full memory continuum (first-author paper, *Geophysical Research Letters*, 2026).
- Embedded Liang-Kleeman information flow: derived the directed flow in closed form with its white-noise and frozen-noise limits; a time-resolved ensemble estimator tracks theory while the reverse flow stays at zero.
- Reconciled maximum-entropy theory and Langevin dynamics through the choice of size variable, generalized the model to any conserved moment order, and proved a no-go result for boundary rules at zero size.
- Derived exact stationary distributions of a Fokker-Planck model with re-injection boundaries; solved the 2D hypoelliptic problem by conservative finite volumes (upwind, Chang-Cooper) with a Woodbury low-rank solve.
- Compared absorbing, reflecting and re-injection boundary conditions at zero droplet size; convergence studies in time step and ensemble size showed the boundary choice sets the relative dispersion more than any physical parameter.

Graduate Teaching Assistant, AMS 595 Fundamentals of Computing (Python, C++, MATLAB), Stony Brook University Aug 2026 – Present

CAUSAL GRAPH LEARNING AND CAUSAL ML (WORKING DRAFTS)

- **Entropy-integrated causal graph learning.** Goal: replace correlation and Gaussian statistics with entropy evidence at every stage: transfer entropy initialization, conditional mutual information tests (PC, PCMCI), entropy-regularized NOTEARS and DYNOTEARS, and doubly robust effect estimation (DML, AIPW, TMLE).
- **Pretrained causal discovery for time series (causal ML).** Goal: a transformer pretrained on simulated systems with known causal graphs that returns a time-series PAG with a probability on every edge mark.

PUBLICATIONS AND PRESENTATIONS

- **Shu, Y., Liu, Y., Zhang, T., McGraw, R., Li, X. (2026).** Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, 53(17), e2026GL123056.
- In preparation: *Physical Review E* (maximum entropy); *J. Atmos. Sci.* (entrainment-mixing); causal discovery methodology.
- Presentations: AGU Fall Meeting 2025 (first-author poster, A21B-2034); 106th AMS Annual Meeting 2026 (co-author).

SKILLS

Causal inference and information theory: Liang-Kleeman information flow, transfer entropy, nearest-neighbor conditional mutual information, path-space relative entropy, PAG search (FCI, LPCMCI, PCMCI+), surrogate tests

Stochastic modeling and numerics: Itô calculus, Langevin and Fokker-Planck equations, Ornstein-Uhlenbeck processes, Girsanov's theorem, maximum entropy, stochastic differential equation integrators, Monte Carlo, finite-volume PDE solvers

Statistics and ML: time-series analysis, change-point detection, PCA and ICA, false-discovery control, gradient boosting

Programming: Python (NumPy, SciPy, pandas, scikit-learn, statsmodels), C++, MATLAB, R; Git, pytest, Linux, SLURM/HPC

Code: github.com/syaohui